

A linear equation

A linear equation in one variable can be written with real constants a and b , where a is nonzero.

$$ax + b = 0$$

What makes this equation linear?

A solution

A solution is a value that makes the original equation true.

$$4x - 7 = 9$$

Is $x = 4$ a solution?

Equivalent equations

Equivalent equations have exactly the same solution set.

Which operations preserve an equation's solutions?

Solving with distribution

$$3(2x - 5) = 2x + 9$$

What is the solution?

Solving with fractions

$$\frac{x-2}{3} + \frac{x}{2} = 4$$

What is the solution?

An identity

An identity is true for every allowed value of the variable.

$$2(x + 3) = 2x + 6$$

What is its solution set?

An inconsistent equation

An inconsistent equation has no solution.

$$4x + 5 = 4x - 2$$

What contradiction shows that the solution set is empty?

Keep possible zero solutions

$$x(x - 3) = 0$$

Would dividing both sides by x preserve every solution?

A rational equation

$$\frac{x+1}{x-2} = 3$$

Which values are excluded, and what solution remains?

Combine before solving

$$\frac{2}{x} + \frac{1}{x} = \frac{3}{5}$$

After combining the left side, what is the solution?

A rejected candidate

$$\frac{x+2}{x-3} = \frac{5}{x-3}$$

Does simplifying this equation produce an allowed solution?

Rearrange a formula

The width w is positive.

$$A = \ell w$$

How can we express the length ℓ in terms of A and w ?

Slope

For two points with different x-coordinates, slope is the ratio of vertical change to horizontal change.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

What happens when the two x-coordinates are equal?

Slope from two points

$(1, 5), \quad (5, -3)$

What is the slope of the line through these points?

Slope-intercept form

$$y = mx + b$$

What do m and b tell us about the graph?

Read a line's equation

$$3x + 2y = 8$$

What are this line's slope and y-intercept?

Point-slope form

A line has slope m and passes through (x_1, y_1) .

$$y - y_1 = m(x - x_1)$$

Why does the given point satisfy this equation?

A line through two points

$(-2, 1), \quad (4, 10)$

What equation describes the line through these points?

Horizontal and vertical lines

A line passes through $(3, -4)$.

What equation describes the horizontal line through this point?

A vertical line

A line passes through $(3, -4)$.

What equation describes the vertical line through this point?

Parallel lines

Distinct nonvertical lines are parallel when their slopes are equal.

$$y = 3x + 2, \quad y = 3x - 5$$

How can their equations show that they never intersect?

A parallel line

$$y = -2x + 6$$

What equation describes the parallel line through (4, 1)?

Perpendicular lines

Nonvertical perpendicular lines have slopes whose product is -1.

$$m_1 m_2 = -1$$

What kind of line is perpendicular to a horizontal line?

A perpendicular line

$$y = \frac{2}{3}x - 1$$

What equation describes the perpendicular line through (2, 5)?